RV College of Engineering®, Bengaluru - 59

Department of Information Science and Engineering

Database Management Systems (CD252IA)

SYNOPSIS

TITLE		
CHESS TOURNAMENT MANAGEMENT AND PLAYER PERFORMANCE ANALYTICS SYSTEM		
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1. Introduction

Running a chess tournament requires player registration, round scheduling, accurate result entry, and repeated recalculation of standings. When handled manually or through simple spreadsheets, records can become inconsistent and long-term player performance is difficult to analyse.

The Chess Tournament Management and Player Performance Analytics System is a database-backed application that handles the complete tournament lifecycle, including player registration, tournament creation, round and match scheduling, result recording, rankings, and rating updates. It also analyses rating changes, opponent strength, and notable upset victories.

The relational database supports Elo-style rating calculation, rating history, opponent-strength comparison, win/loss/draw statistics, rating-progression charts, and automatic upset detection.

2. Existing System

Existing tournament tools mainly support administration such as player registration, tournament creation, match scheduling, result entry, and leaderboard generation.

They generally provide limited long-term performance analysis. Rating progression across tournaments, opponent-strength comparison, and automatic identification of upset victories are often unavailable and must be analysed separately.

3. Proposed System

The proposed system combines tournament administration and player-performance analytics in one platform.Key features include: Player registration and profile management; Tournament creation and management; Registering players into a specific tournament; Creating rounds and

assigning matches within them; Recording game results; Automatic calculation of points and standings; Elo-style rating updates after every game; A running history of rating changes per player; Player performance analysis across tournaments; Opponent-strength analysis; Detection of upset victories; A dashboard summarising tournament and player analytics.

Novelty: The novelty is its analytics layer. For every game, the system compares player ratings, calculates the expected outcome and rating change, stores rating progression, and automatically identifies meaningful upset victories when a lower-rated player defeats a significantly higher-rated opponent.

4. Relational Database Structure

The relational schema contains Player, Tournament, Registration, Round, Game, Result, Rating History, and Tournament Standings.

Primary and foreign keys, constraints, and normalisation maintain consistency. Views, triggers, stored procedures, transactions, aggregate queries, joins, indexing, and window functions support reporting and analytics.

5. RDBMS

The system uses PostgreSQL or MySQL to store and manage players, tournaments, registrations, rounds, games, results, ratings, and standings. Primary keys, foreign keys, constraints, and normalization ensure data integrity and consistency. Joins, views, triggers, stored procedures, and transactions support efficient tournament management, rating calculations, and player-performance analytics.

6. Societal Concern

Chess clubs, schools, and independent organisers often rely on manual methods that can cause inconsistent records and disputes. Centralising player data, match history, results, and performance records improves transparency, reduces administrative effort, and gives players reliable feedback on their progress.

7. Innovative Experiment

A machine learning model will use historical player, game, result, and rating data to predict player performance and store the generated predictions in the database for further analysis. The model will continuously learn from newly recorded tournament results and rating changes to improve the accuracy of future player-performance predictions.